Ttukttak Vibe Workshop: New Features Lesson Supplement

Read alongside the Teacher guide (14 pp.) · Screens as of Nov 3, 2026

How to use the features added to the app after the Teacher's Guide was written. Keep the 8-lesson flow as is and add these where they fit. No sign-up is needed; everything is stored only in the browser on the student's device and leaves it only when files are exchanged.

Feature	Best lesson	What students do
4 pattern-drawing missions	Lessons 3–5 or extra	Color cells with commands, loops and conditions; compare with target
Easy mode (grades 3–4)	All lessons	Build requests with word blocks, read aloud, large text
Peer check files	Lesson 7	Exchange request files to check a friend's work and merge results
Self-assessment	Lesson 8	Choose own level on four observed items and write what to try next
Lesson presentation screen	Teacher · all lessons	Show stages, activity timer and partner role swaps on the projector
Undo · Redo	All lessons	Undo mistaken edits and version restores one step at a time
English display	As needed	View student screens in English and hear read-aloud in English
Tailored hints	Lessons 3–5	When auto-check fails, see the problems the app found and hints
Export as app file	Lesson 8	Save the app as one file to show family and friends
QR exchange	Lesson 7	Send works and peer check requests to other devices by QR
Display	As needed	Turn on high contrast, readable font and wide spacing
Teacher's missions · class graph	Teacher	Make missions for your class; view class progress as a graph
End-of-class wrap-up	End of every lesson	Back up → Start next user or delete all, in order
If-then rule app (4 missions)	Lessons 4–5 or extra	Change values with buttons; check condition order and boundaries
Endless bug detective	Early finishers	Find and fix a hidden bug; confirm with auto-check
My mission map · Growth portfolio	Lesson 8 · end of term	See missions completed; print records from several works at once
Work showcase · Lesson kit	Teacher	Present works in turn; print the lesson kit on one A4 sheet

1. Pattern-drawing missions

Students run commands (direction · steps) in order on a grid to color cells. Four missions follow in sequence: ‘Commands move in order’ (sequence) → ‘Repeat it and you get stairs’ (repetition) → ‘Huh? The drawing comes out the other side!’ (practice bug) → ‘Follow step by step to finish the square’ (find the missing command).

- Predict first: have students read the commands and pre-color the shape they expect on the worksheet grid.
- Use the ‘Draw one step’ button on the run screen to follow one step at a time and read ‘what the app remembers now’ (current position · colored cells · wrapped · stopped).
- Mission 3 contains a practice bug: leaving the board wraps to the opposite side. In Settings, change ‘Board edge rule’ to ‘Stop at the edge (normal rule)’ and check again.
- Go further: under ‘Condition rule’ in Settings, choose ‘Turn right and keep going’ so the pen turns right when the next cell is blocked (board edge · already colored). Have students draw a border with one long command.

- Turn on ‘Show target as dotted’ to compare against the dotted cells. The app reports how many cells are missing and how many are extra.

Sample questions

- Why does repeating the same command three times make a staircase?
- What should the app check the moment it leaves the board?
- Why can one command plus the ‘turn right when blocked’ condition draw a whole border?

2. Easy mode (grades 3–4)

Choose ‘Easy mode (grades 3–4)’ under ‘Guide mode’ at the top of the screen. Only the display changes; records stay the same.

- Word blocks: tap the ‘Who uses it? → What will we make? → What happens when pressed? → Rules to keep’ blocks to build the request text. Tap again to remove.
- Prediction starter blocks: continue a prediction with ‘If I press this / I think this will happen. / Because’.
- Read aloud: step guidance, mission text and the sentences students make are read with the device's voice. If the screen language is English, an English voice is used. If there is no sound, check the device volume and the browser's speech support.
- Less writing is fine. Let students explain predictions and observations aloud, and have the teacher write down just one representative sentence.

3. Exchanging peer check files

Peer checks use two files, with no server or student accounts. The request file contains only the app settings and check items, no other learning records.

Order	Who	What to do on screen
①	Student who made it	Download the ‘Peer check request file’ in step 4 or send it via the share dialog.
②	Friend checking	My work shelf → open the request file with ‘Import file’. Predict first, press it yourself, then record the actual result, verdict and comments for each check.
③	Friend checking	Make a ‘Check result file’ and return it.
④	Student who made it	Open the result file with ‘Import file’ and it is added as a ‘Friend · teacher comments’ check record.

- If the work was moved to another device and the original cannot be found, a screen appears for choosing which work to attach the result to. Works from the same mission are listed first.
- Up to 100 check records per work. If the limit is exceeded, remove unneeded records and import again.
- A friend's comments are not the app's verdict. Have students also read the reasoning that compares expected and actual results.

4. Self-assessment and class assessment

- In step 5, ‘Explain and save’, students choose their own level on the four process rubric items (problem and requirements · understanding how it works · checking and fixing · explaining and applying to new tasks) and write ‘what to try next’.
- The portfolio's ‘Process assessment · me and my teacher’ table shows self-assessment and teacher assessment side by side. These are not scores computed by the app.
- Teachers gather student work files in the ‘Collect class work’ tab of the Teacher Room, review the level table, and return assessments as a file.

- The class list being assessed stays open on the same device after a refresh or reopen. Use ‘Clear list’ when the lesson is over.

5. Lesson presentation screen

Open ‘Lesson presentation screen’ from a lesson in the Teacher Room. It is designed for a projector.

- Lesson stages: shows Intro · Activity · Wrap-up with today's mission, predict first, what to check, and closing questions in order.
- Activity timer: set a time and start/stop. When it ends, ‘Activity time is up.’ appears in large text.
- Partner role swap: every 10 minutes it prompts swapping the driver (partner A · partner B) and observer.
- Sound alerts: with ‘Also alert with sound’ on, it chimes twice at a swap and three times at the end. Sound is prepared on the first ‘Start’, so check the classroom speaker volume beforehand.
- Full screen: supported browsers show a ‘Full screen’ button. On unsupported devices (some tablets · phones), it explains how to enlarge via the browser menu instead.

6. Undo and redo

- Use the ‘Undo’ · ‘Redo’ buttons above the work screen. Outside text fields, Ctrl+Z · Ctrl+Shift+Z (or Ctrl+Y; ⌘ on Mac) also work. Inside a text field, the browser's own text undo runs first. Consecutive keystrokes are grouped into one step.
- Saving, restoring and deleting versions, and deleting check records, can also be undone. Button presses on the run screen are not added to undo history.
- Undo history keeps up to 50 steps per work, in the same browser tab only. It survives a refresh but is lost when the tab closes. Use ‘Save version’ for anything to keep long-term.
- ‘Start next user’ or ‘Delete all app data’ also clears undo history.

7. English display

- Choose English under ‘언어 · Language’ at the top of the screen. The entire student interface (home · first try · missions · the five work steps · settings · checks · action log · portfolio · work shelf · peer check · word blocks · error messages) and mission text switch to English.
- Teacher screens (lesson guides · Collect class work · Lesson presentation screen · Make a mission · Work showcase · lesson kit), finished examples and mission starter text also switch to English. Printable teacher materials (PDF · Word) and teacher-made missions stay in Korean. English worksheets for students are available separately.
- Student writing and saved version names do not change. Records stay the same when the language is switched.

8. Cautions for shared devices

- Records are stored in the device's browser, not in a student account. The next person using the device can see them, so teach students not to enter personal information.
- At the end of the lesson, download and keep the work files, then press ‘Start next user’. Previous records are collapsed in the list and do not carry over automatically (this is not privacy protection).
- When returning or resetting a device, use ‘Delete all app data’. Data from other apps at the same address is not deleted.
- The ‘End-of-class wrap-up’ on the work shelf walks through: confirm saves → download all works as files → Start next user or delete all. The final button is enabled only after choosing back up or skip.

9. Tailored hints

- In step 4, press ‘Auto-check current settings’; if any check fails, ‘Problems the app found and tailored hints’ appears. Only the problem is shown at first; press ‘Show hint’ to reveal how to fix it.
- Hints come from the same rules as auto-check, not AI. Examples: pressing the same correct answer repeatedly raises the score again → scoring rule; the target ending is unreachable → scene link map; a missing cell in a pattern → the command from the last position to the nearest missing cell.
- To have students find practice bugs themselves, have them record predictions and observations before auto-check. Even after seeing a hint, they should record what happened when they pressed it themselves.

10. Export as app file and QR exchange

- ‘Export as app file’ on the work screen and portfolio saves the app as a single HTML file. It opens in a browser offline and contains only the work settings, no learning records.
- ‘Send by QR’ (work) and ‘Send request by QR’ (peer check) display a large QR code. Point the receiving device's camera app at it to open the link, then press ‘Receive as new work’ or ‘Open in peer check’.
- The content after # in the QR link is not sent to a server. Large works, such as stories with many scenes, do not fit in a QR code; tell students to send a file instead.

11. Display (high contrast · readable font)

- Turn on ‘High contrast’ under ‘Display’ at the top of the screen: text and borders become black, the background white, and selected buttons fill black.
- ‘Readable font · wide spacing’ enlarges text and widens letter, word and line spacing. Recommend it for students with low vision or reading difficulties.
- In pattern drawing, target cells can be moved with arrow keys and toggled with Enter or Space.

12. Making teacher's missions and class progress graph

- In ‘Make a mission’ in the Teacher Room, set the project type, title · goal · prediction question · task, starting settings (preview on the run screen), whether to include a practice bug, check items (1–6), hints (1–3), and closing questions.
- Press ‘Add mission to this device’ and it appears in the Mission Workbench as a ‘teacher's mission’. Download it with ‘Download mission file’ and open it on student devices with ‘Import file’ to add it there too.
- To receive works made from a teacher's mission in Collect class work, the teacher's device must have the same mission. Keep the mission file.
- The ‘Class progress summary’ in Collect class work shows how many students are at each next step, how many mission hints were opened, and the average gap between self- and teacher assessment. It gives a one-line note on the step where most students stopped and the item with the largest gap; ‘View as table’ shows the same data as a table.

13. If-then rule app

- The fifth app type. Pressing a button changes a value the app remembers (e.g. sticker count), and the message of the first matching condition from the top (at least · at most · equals) appears. If none match, the ‘otherwise’ message appears.
- The currently matching condition is shown below the run screen, so students see that ‘conditions are checked in order from the top’.
- 4 missions: Celebrate at 5 (condition); No going below 0 (smallest value); The top message never shows (practice bug: a broad condition hides a narrow one); Only when exact (equals vs. at least).

- Sample questions: 6 matches both ‘3 or more’ and ‘6 or more’, so why does only ‘Good’ appear? If you press +2, can you skip 10?
- Worksheet: use “Extra lesson If-then rule app” at the end of the student worksheet set (2 pages, pp. 17–18). We suggest adding one lesson after Lesson 4 (What the app remembers) or Lesson 5 (Bug detective). Start with the mission “Huh? The best message never shows!”; students who finish early go on to “Only when it's exactly right”.
- 40-minute flow: Opening 5 min (show an app that celebrates at 5 stickers and ask “what does the app remember?”) → Activity 30 min (predict 7 min, check the message at 3, 5 and 6 stickers 8 min, fix the condition order and re-check 15 min) → Wrap-up 5 min (explain “only the first matching condition is used” in your own words).

14. Endless bug detective · My mission map · Growth portfolio

- ‘Endless bug detective’ in the Mission Workbench gives a new challenge: a working work with one hidden practice bug. It does not say where the bug is; once fixed, auto-check confirms it and the ‘bugs found and fixed’ count rises. Recommend it for students who finish missions early.
- ‘My mission map’ links missions into paths by type and marks where all five steps are recorded (✓), in progress (…), and not yet (○). There are no rankings or scores.
- ‘End-of-term growth portfolio’ on the work shelf prints (saves as PDF) predictions, reasons for fixes, lessons learned and teacher comments from several works at once. It shows self-assessments from the first and last works side by side, useful for parent conferences and end-of-term assessment.

15. Work showcase · Today's lesson kit · Personal information check

- ‘Work showcase’ in Collect class work runs the collected works one by one on the projector, showing prediction vs. actual, reasons for fixes and how it works, one page each. Use arrow keys to move and Esc to exit. ‘Print praise & question cards’ prints cards for friends to fill in by hand.
- ‘Print one-page lesson kit’ in each Teacher Room lesson puts the lesson goal, 40-minute sequence, mission QR, materials and things to watch for on one A4 sheet. Post the QR on the board and students open the mission directly with their device camera.
- Before sending by file, share, QR or app file, a ‘Check before sending’ dialog appears if text looks like a phone number, email, address, school class or real name. Detection is rule-based and can miss things, so keep reminding students not to enter personal information.
- Works made from a teacher's mission carry the mission along in files, peer checks and QR, so they open even on devices without the mission.

Peer check record sheet

Name: _____ Friend's work checked: _____ Date: _____

1. Predict before pressing

I think this work will behave like this:

_____

_____

_____

2. Press it yourself and record

What to check	Expected result	Actual result	Verdict (○ / to fix)

3. Notes for your friend

What I understood:

_____

What was confusing or could be improved:

_____

4. Did you make a 'Check result file' in the app and return it to your friend? ☐ Yes ☐ Not yet